SHUAIKE SHEN

Carnegie Mellon University, Pittsburgh, PA, United States, 15213
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Education

Carnegie Mellon University — Pittsburgh, PA, U.S.
Ph.D in Computational Biology — *Aug. 2025 - Present*

Zhejiang University — Hangzhou, Zhejiang, China
B.Eng in Computer Science and Technology (Honors) — *Sept. 2021 - June. 2025*

- Mixed Class, Chu Kochen Honors College, GPA: 3.83/4.00

Research Interest

I am deeply interested in developing AI and machine learning methods for scientific discovery, particularly in biomedical applications. My research focuses on protein design and flow-matching models. I have also worked on topics related to scientific large language models, reinforcement learning, and crystal structure modeling.

Research Experience

Research Assistant — **Carnegie Mellon University**
Advisor: Prof. Jian Ma — *Nov. 2025 - Present*

- Working on advanced AI agents to accelerate scientific discovery, with a specialized focus on genomic and cellular analysis.

Research Assistant — **Carnegie Mellon University**
Advisor: Prof. Olexandr Isayev & Prof. Lei Li — *Aug. 2025 - Nov. 2025*

- Working on low energy conformer generation and protein complex using diffusion models.

Research Intern — **Shanghai AI Lab**
Advisor: Dr. Biqing Qi & Dr. Yuqiang Li — *May. 2025 - Aug. 2025*

- Working on large language models for molecular spectra and developing molecular foundation models to bridge spectral data with molecular properties and structures.

Research Intern — **University of Illinois Urbana-Champaign**
Advisor: Prof. Ge Liu & Prof. Nicholas Ching Hai Wu — *June. 2024 - Mar. 2025*

- Working on latent flow matching model for protein sequence and strucutre co-design and developed online reinforcement algorithms for flow matching models.

Research Assistant — **Zhejiang University**
Advisor: Prof. Chunhua Shen & Prof. Hao Chen — *Oct. 2022 - May. 2024*

- Working on protein engineering and crystal structure modeling. I developed vector field protein encoder, multi-motif scaffolding method, RNA inverse folding paradiagm, binding energy prediction model and crystal denoising pretraining framework.

Publications

* denotes equal contribution and † denotes corresponding author.

SkillFoundry: Building Self-Evolving Agent Skill Libraries from Heterogeneous Scientific Resources — Apr. 2026

- **Shuaike Shen***, Wenduo Cheng*, Mingqian Ma, Alistair Turcan, Martin Jinye Zhang, Jian Ma†
- **Under review**, Paper, Code

MolAct: An Agentic RL Framework for Molecular Editing and Property Optimization — Dec. 2025

- Zhuo Yang*, Yeyun Chen*, Jiaqing Xie, Ben Gao, **Shuaike Shen**, Wanhao Liu, Liujia Yang, Beilun Wang, Tianfan Fu, Yuqiang Li†
- **Technical Report**, Paper, Code

R^2Seg: Training-Free OOD Medical Tumor Segmentation via Anatomical Reasoning and Statistical Rejection Nov. 2025

- **Shuaike Shen***, Ke Liu*, Jiaqing Xie, Shangde Gao, Chunhua Shen, Ge Liu, Mireia Crispin-Ortuzar, Shangqi Gao††
- **CVPR 2026 Oral**, Paper, Code

SpecMol: A Spectroscopy-Grounded Foundation Model for Multi-Task Molecular Learning Sep. 2025

- **Shuaike Shen**, Jiaqing Xie, Zhuo Yang, Antong Zhang, Shuzhou Sun, Ben Gao, Tianfan Fu†, Biqing Qi†, Yuqiang Li†
- **Under review**, Paper, Code

SpectrumWorld: Artificial Intelligence Foundation for Spectroscopy Aug. 2025

- Zhuo Yang*, Jiaqing Xie*, **Shuaike Shen**, Daolang Wang, Yeyun Chen, Ben Gao, Shuzhou Sun, Biqing Qi, Dongzhan Zhou, Lei Bai, Linjiang Chen, Shufei Zhang, Jun Jiang†, Tianfan Fu†, Yuqiang Li†
- **Under review**, Paper

Intern-S1: A Scientific Multimodal Foundation Model Aug. 2025

- **Intern-S1 team**
- **Technical Report**, Paper, Code, Huggingface

From Sentences to Sequences: Rethinking Languages in Biological System July. 2025

- Ke Liu*, **Shuaike Shen***, Hao Chen†
- Paper, Code

Online Reward-Weighted Fine-Tuning of Flow Matching with Wasserstein Regularization Jan. 2025

- Jiajun Fan, **Shuaike Shen**, Chaoran Cheng, Yuxin Chen, Chumeng Liang, Ge Liu†
- **ICLR 2025**, Paper

A Denoising Pre-training Framework for Accelerating Novel Material Discovery Dec. 2024

- **Shuaike Shen**, Ke Liu, Hao Chen†
- **AAAI 2025**, Paper

Training Text-to-Image Flow Models on Self-Generated Data Nov. 2024

- Jiajun Fan, **Shuaike Shen**, Chaoran Cheng, Chumeng Liang, Ge Liu†
- **Under review**

Physics-Informed Neural Networks for Unsupervised Binding Energy Prediction Aug. 2024

- Ke Liu*, **Shuaike Shen***, Hao Chen†
- **Under review**, Manuscript

Boost Your Crystal Model with Denoising Pre-training June. 2024

- **Shuaike Shen***, Ke Liu*, Muzhi Zhu, Hao Chen†
- **ICML 2024 AI for Science Workshop**, Paper, Project Page

FADiff: Floating Anchor Diffusion Model for Multi-motif Scaffolding May. 2024

- Ke Liu*, **Shuaike Shen***, Weian Mao*, Xiaoran Jiao, Zheng Sun, Hao Chen†, Chunhua Shen†
- **ICML 2024**, Paper, Project Page

De novo protein design using geometric vector field networks Oct. 2023

- Weian Mao*, Muzhi Zhu*, Zheng Sun*, **Shuaike Shen**, Lin Yuanbo Wu, Hao Chen†, Chunhua Shen†
- **ICLR 2024 spotlight**, Paper, Project Page

Selected Projects

VITON: High resolution virtual try on based on generative models 2023
Scientific research projects of Chu Kochen Honors College advised by Prof. Hao Chen

MiniSQL: A database system implemented in C++ 2023
The course project of Database systems and concepts

Parallel optimization of big data pivots selection based on vectorization and OpenMP 2022
High Performance Computing 101 summer course | ACM-China IPCC 2022

Skills

Programming Skills: Python, C/C++, PyTorch, SQL

Language Skills: Mandarin (Native), English (TOEFL 104)